CS 61C: Great Ideas in Computer Architecture (Machine Structures)
Caches Part 2

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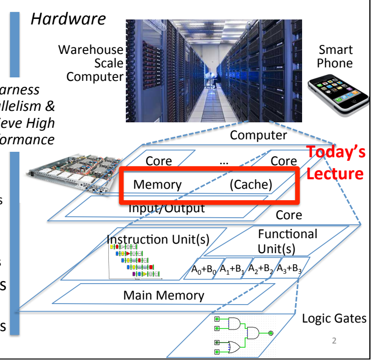
http://inst.eecs.berkeley.edu/~cs61c/

You Are Here!

Software

- Parallel Requests
  Assigned to computer
  e.g., Search "Katz"
- Parallel Threads
  Assigned to core
  e.g., Lookup, Ads
- Parallel Instructions
  >1 instruction @ one time
  e.g., 5 pipelined instructions
- Parallel Data
  >1 data item @ one time
  e.g., Add of 4 pairs of words
- Hardware descriptions
  All gates @ one time
- Programming Languages



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Caches Review

- Principle of Locality
  - Temporal Locality and Spatial Locality
- Hierarchy of Memories (speed/size/cost per bit) to Exploit Locality
- Cache – copy of data in lower level of memory hierarchy
- Direct Mapped to find block in cache using Tag field and Valid bit for Hit
- Cache design choice:
  - Write-Through vs. Write-Back

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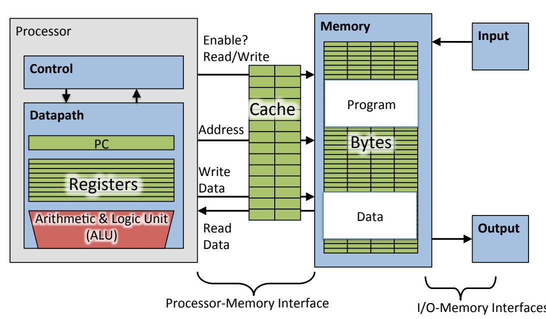
Review: Adding Cache to Computer



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Processor Address Fields used by Cache Controller

- Block Offset: Byte address within block
- Set Index: Selects which set
- Tag: Remaining portion of processor address

Processor Address (32-bits total)

Tag	Set Index	Block offset

- Size of Index = log2 (number of sets)
- Size of Tag = Address size – Size of Index – log2 (number of bytes/block)

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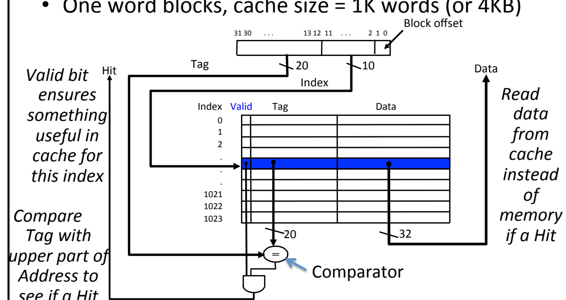
Review: Direct-Mapped Cache

- One word blocks, cache size = 1K words (or 4KB)



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Cache Terms

- Hit rate: fraction of accesses that hit in the cache
- Miss rate: 1 – Hit rate
- Miss penalty: time to replace a block from lower level in memory hierarchy to cache
- Hit time: time to access cache memory (including tag comparison)

- Abbreviation: “$” = cache (A Berkeley innovation!)

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Average Memory Access Time (AMAT)

- Average Memory Access Time (AMAT) is the average to access memory considering both hits and misses in the cache

AMAT = Time for a hit
+ Miss rate x Miss penalty

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Clickers/Peer instruction

AMAT = Time for a hit + Miss rate x Miss penalty

Given a 200 psec clock, a miss penalty of 50 clock cycles, a miss rate of 0.02 misses per instruction and a cache hit time of 1 clock cycle, what is AMAT?

☐ A: ≤200 psec

☐ B: 400 psec

☐ C: 600 psec

☐ D: ≥ 800 psec

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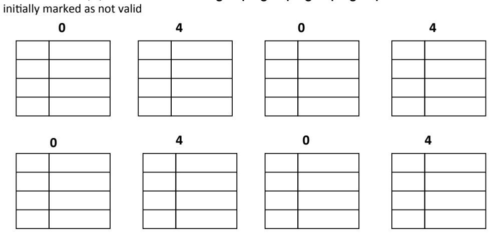
Example: Direct-Mapped $ with 4 Single-Word Lines, Worst-Case Reference String

- Consider the main memory address reference string

Start with an empty cache - all blocks initially marked as not valid 0 4 0 4 0 4 0 4



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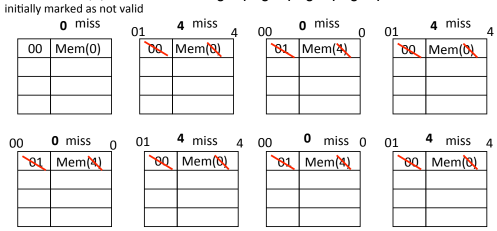

- 8 requests, 8 misses
- Ping-pong effect due to conflict misses - two memory locations that map into the same cache block

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Alternative Block Placement Schemes

- DM placement: mem block 12 in 8 block cache: only one cache block where mem block 12 can be found—(12 modulo 8) = 4
- SA placement: four sets x 2-ways (8 cache blocks), memory block 12 in set (12 mod 4) = 0; either element of the set
- FA placement: mem block 12 can appear in any cache blocks

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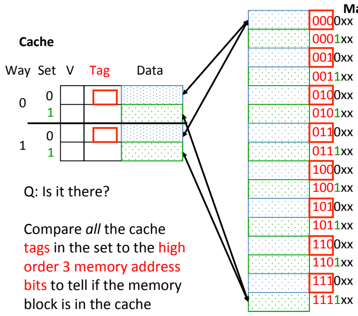
Example: 2-Way Set Associative $
(4 words = 2 sets x 2 ways per set)



One word blocks
Two low order bits define the byte in the word (32b words)

Q: Is it there?

Compare *all* the cache tags in the set to the high order 3 memory address bits to tell if the memory block is in the cache

Q: How do we find it?

Use next 1 low order memory address bit to determine which cache set (i.e., modulo the number of sets in the cache)

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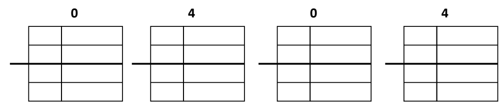
Example: 4 Word 2-Way SA $
Same Reference String

- Consider the main memory word reference string

Start with an empty cache - all blocks initially marked as not valid 0 4 0 4 0 4 0 4



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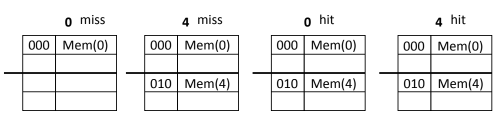
Example: 4-Word 2-Way SA $
Same Reference String

- Consider the main memory address reference string

Start with an empty cache - all blocks initially marked as not valid 0 4 0 4 0 4 0 4



- 8 requests, 2 misses
- Solves the ping-pong effect in a direct-mapped cache due to conflict misses since now two memory locations that map into the same cache set can co-exist!

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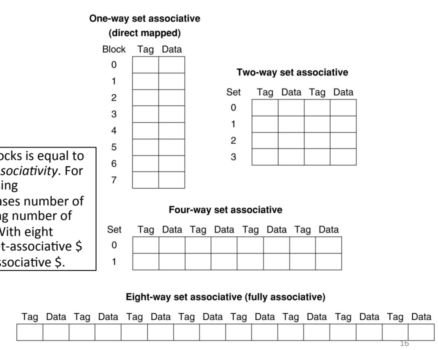
Different Organizations of an Eight-Block Cache

Total size of $ in blocks is equal to *number of sets x associativity*. For fixed $ size, increasing associativity decreases number of sets while increasing number of elements per set. With eight blocks, an 8-way set-associative $ is same as a fully associative $.



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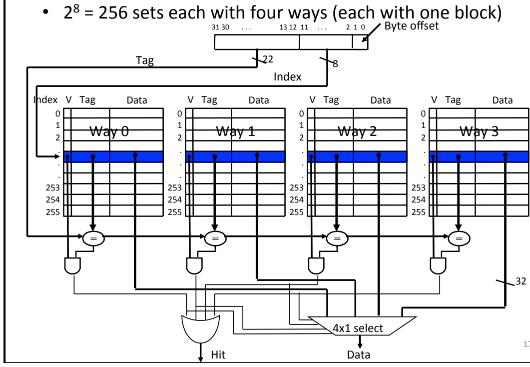
Four-Way Set-Associative Cache

- 2^8 = 256 sets each with four ways (each with one block)



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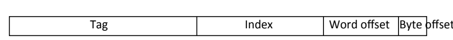
Range of Set-Associative Caches

- For a fixed-size cache, each increase by a factor of two in associativity doubles the number of blocks per set (i.e., the number or ways) and halves the number of sets – decreases the size of the index by 1 bit and increases the size of the tag by 1 bit



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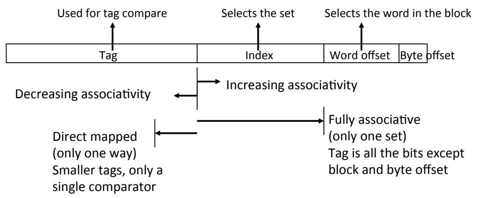
Range of Set-Associative Caches

- For a *fixed-size* cache, each increase by a factor of two in associativity doubles the number of blocks per set (i.e., the number or ways) and halves the number of sets – decreases the size of the index by 1 bit and increases the size of the tag by 1 bit

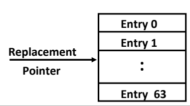


Costs of Set-Associative Caches

- N-way set-associative cache costs
  - N comparators (delay and area)
  - MUX delay (set selection) before data is available
  - Data available after set selection (and Hit/Miss decision). DM $: block is available before the Hit/Miss decision
    - In Set-Associative, not possible to just assume a hit and continue and recover later if it was a miss
- When miss occurs, which way's block selected for replacement?
  - Least Recently Used (LRU): one that has been unused the longest (principle of temporal locality)
    - Must track when each way's block was used relative to other blocks in the set
    - For 2-way SA $, one bit per set → set to 1 when a block is referenced; reset the other way's bit (i.e., "last used")

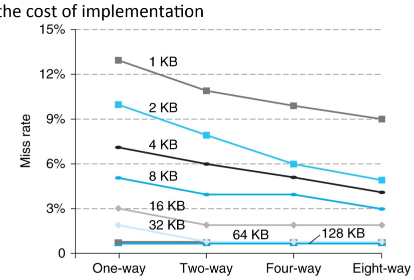
Cache Replacement Policies

- Random Replacement
  - Hardware randomly selects a cache evict
- Least-Recently Used
  - Hardware keeps track of access history
  - Replace the entry that has not been used for the longest time
  - For 2-way set-associative cache, need one bit for LRU replacement
- Example of a Simple "Pseudo" LRU Implementation
  - Assume 64 Fully Associative entries
  - Hardware replacement pointer points to one cache entry
  - Whenever access is made to the entry the pointer points to:
    - Move the pointer to the next entry
  - Otherwise: do not move the pointer
  - (example of "not-most-recently used" replacement policy)



Benefits of Set-Associative Caches

- Choice of DM $ or SA $ depends on the cost of a miss versus the cost of implementation



- Largest gains are in going from direct mapped to 2-way (20%+ reduction in miss rate)

Administrivia

- Project 2-1 due Sunday March 15th, 11:59PM
  - Use pinned Piazza threads!
  - We'll penalize those who ask, but don't search!
- Guerilla sections starting this weekend
  - Optional sections, focus on lecture/exam material, not projects
  - Vote for time slot on Piazza poll

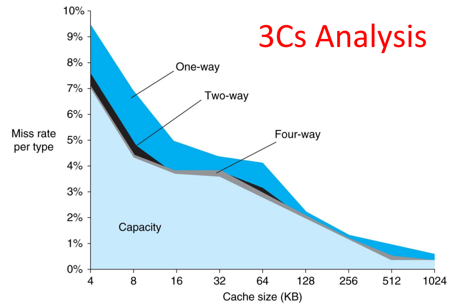
Understanding Cache Misses: The 3Cs

- Compulsory (cold start or process migration, 1st reference):
  - First access to block impossible to avoid; small effect for long running programs
  - Solution: increase block size (increases miss penalty; very large blocks could increase miss rate)
- Capacity:
  - Cache cannot contain all blocks accessed by the program
  - Solution: increase cache size (may increase access time)
- *Conflict (collision):*
  - *Multiple memory locations mapped to the same cache location*
  - *Solution 1: increase cache size*
  - *Solution 2: increase associativity (may increase access time)*

How to Calculate 3C's using Cache Simulator

1. *Compulsory*: set cache size to infinity and fully associative, and count number of misses
2. *Capacity*: Change cache size from infinity, usually in powers of 2, and count misses for each reduction in size
   – 16 MB, 8 MB, 4 MB, … 128 KB, 64 KB, 16 KB
3. *Conflict*: Change from fully associative to n-way set associative while counting misses
   – Fully associative, 16-way, 8-way, 4-way, 2-way, 1-way

3Cs Analysis



- Three sources of misses (SPEC2000 integer and floating-point benchmarks)
  – Compulsory misses 0.006%; not visible
  – Capacity misses, function of cache size
  – Conflict portion depends on associativity and cache size

Improving Cache Performance

AMAT = Time for a hit + Miss rate x Miss penalty

- Reduce the time to hit in the cache
  – E.g., Smaller cache
- Reduce the miss rate
  – E.g., Bigger cache
- Reduce the miss penalty
  – E.g., Use multiple cache levels

Impact of Larger Cache on AMAT?

- 1) Reduces misses (what kind(s)?)
- 2) Longer Access time (Hit time): smaller is faster
  – Increase in hit time will likely add another stage to the pipeline
- At some point, increase in hit time for a larger cache may overcome the improvement in hit rate, yielding a decrease in performance
- Computer architects expend considerable effort optimizing organization of cache hierarchy – big impact on performance and power!

Clickers: Impact of longer cache blocks on misses?

- For fixed total cache capacity and associativity, what is effect of longer blocks on each type of miss:
  – A: Decrease, B: Unchanged, C: Increase
- Compulsory?
- Capacity?
- Conflict?

Clickers: Impact of longer blocks on AMAT

- For fixed total cache capacity and associativity, what is effect of longer blocks on each component of AMAT:
  – A: Decrease, B: Unchanged, C: Increase
- Hit Time?
- Miss Rate?
- Miss Penalty?

Clickers/Peer Instruction:
For fixed capacity and fixed block size, how does increasing associativity effect AMAT?

A: Increases hit time, decreases miss rate

B: Decreases hit time, decreases miss rate

C: Increases hit time, increases miss rate

D: Decreases hit time, increases miss rate

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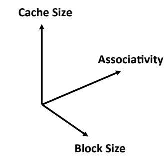
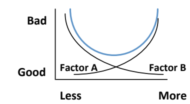
Cache Design Space

- Several interacting dimensions
  - Cache size
  - Block size
  - Associativity
  - Replacement policy
  - Write-through vs. write-back
  - Write allocation
- Optimal choice is a compromise
  - Depends on access characteristics
    - Workload
    - Use (I-cache, D-cache)
  - Depends on technology / cost
- Simplicity often wins





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And, In Conclusion …

- Name of the Game: Reduce AMAT
  - Reduce Hit Time
  - Reduce Miss Rate
  - Reduce Miss Penalty
- Balance cache parameters (Capacity, associativity, block size)

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